Verified square-packing improvements

Exact rational coordinates, replayable checks, and before/after figures

Search on 17 October 2026 | Status: stopped at user request; audit passed

Unit squares	Website-listed side	Certified upper bound	Reduction
227	15.57106781186547	15.563796703670	0.007271108196
229	15.65685424949238	15.650174756949	0.006679492544
260	16.65685424949238	16.645778804182	0.011075445311

What is certified
Each witness gives rational centers and a rational unit basis (c,s), with c^2+s^2=1 exactly. Its four vertices are center +/- (c,s)/2 +/- (-s,c)/2.

Three exact algorithms verify square count, unit geometry, walls and every pair. Two use projection separation; a third uses convex polygon clipping and exact intersection area. Boundary contact is allowed. Decimal bounds round the exact sides upward.

What this does not prove
Optimality, uniqueness, global novelty beyond the checked website corpus, or a formal theorem-prover replay. All three implementations belong to the same AI-assisted campaign and share Fraction arithmetic and rational witness inputs.

Replay
`python3 verify_all.py`
Python 3.10 or later. No third-party packages are required for certificate checks. records.json names the authoritative exact files and their SHA-256 hashes.

Orange topology annotations
The next pages compare actual before/after geometry. Orange marks changed near-contact adjacencies under a documented symmetry/assignment matching. This is an approximate visual layer; exact feasibility is checked separately.

Credit and source
Prepared for JoostdeWinter, with OpenAI Codex-assisted search and verification. Baseline: jlevy.github.io/squares/, checked 17 October 2026. The supplied GitHub draft follows the project's result-registration issue template.

Post-stop audit

Three improved counts in one related construction family

Search stopped early
The planned allocation was 09:20:12-13:20:12 UTC. The user requested stopping at 12:49:15 UTC; all optimization was confirmed stopped by 12:51:59 UTC, about 3 hours 32 minutes into the extension. No numerical search was resumed during this audit.

Geometry and provenance: passed
The frozen 227-, 229- and 260-square witnesses passed all three exact algorithms. Each algorithm considers 85,427 pairs across the three packings. The new clipping checker also passed 12 deliberate corrupted-input controls and seven analytic area tests. Source hashes, square counts, exact geometry, walls and upward decimal rounding agree with the frozen registry.

Visual comparison: passed
Witness hashes, square matching, orange changed-adjacency annotations and projected SVG geometry were audited. Orange uses a near-contact distance of 0.001 unit and a noncanonical square correspondence; it does not assert exact contact.

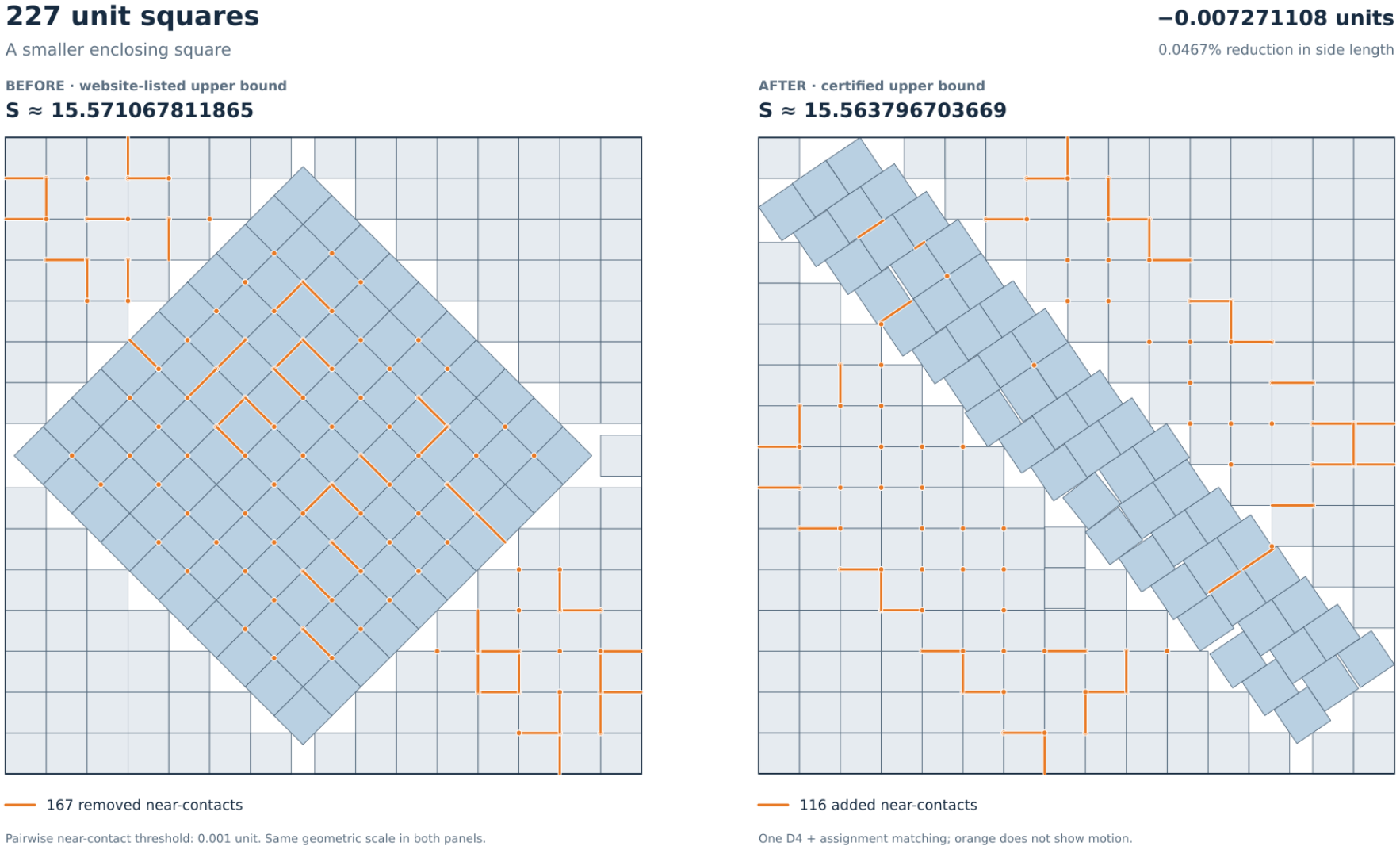
Repairs made
Corrected canceled-run status and duration; excluded mutable Python bytecode from checksums and archives; repaired stale source references for three larger derived packings. These repairs changed no packing geometry. The audit did not find a further promotable improvement in the watched candidate files.

Counting and limitations
227 and 260 arose in the same Young-corner/slipped-ribbon framework; 229 descends from 260 by erosion and refinement. Retired 507/553 claims, larger composites and equal-bound alternatives add no website records. The checked finite public corpus and deletion/framing/product closure do not supersede the three results. This is not an exhaustive worldwide novelty or optimality proof. No GitHub publication or submission has been made.

Detailed machine-readable receipts and audit notes are included in the repository audit/ folder. The original promotion receipts retain two-checker history; the separate audit receipts record the third algorithm.

227 unit squares: before and after

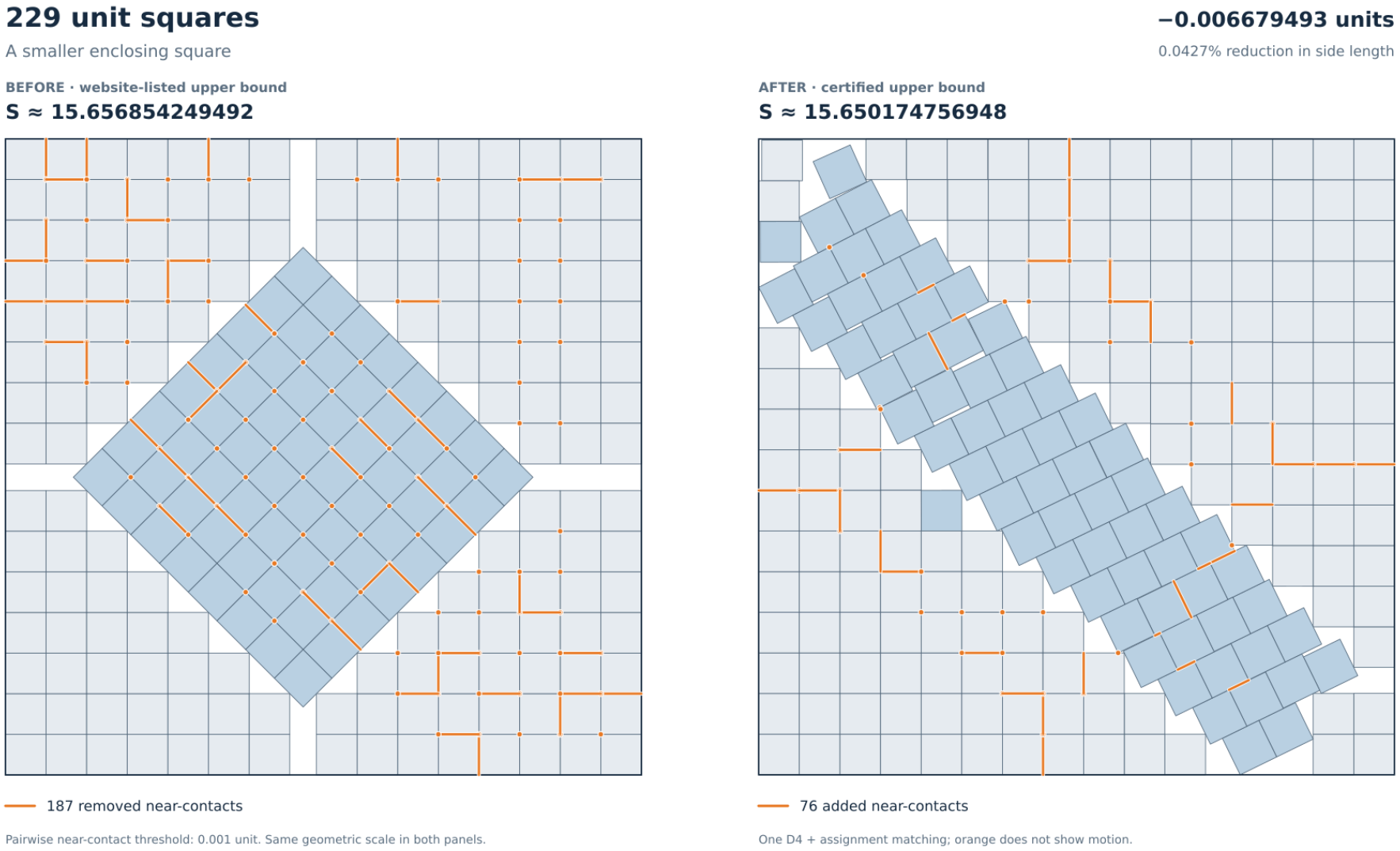
Website 15.571067811865 -> Certified upper bound 15.563796703670



Orange identifies changed near-contact adjacencies after the recorded square-symmetry and tile matching. The accompanying topology JSON states the tolerance and correspondence; the exact certificate, not the figure, establishes feasibility.

229 unit squares: before and after

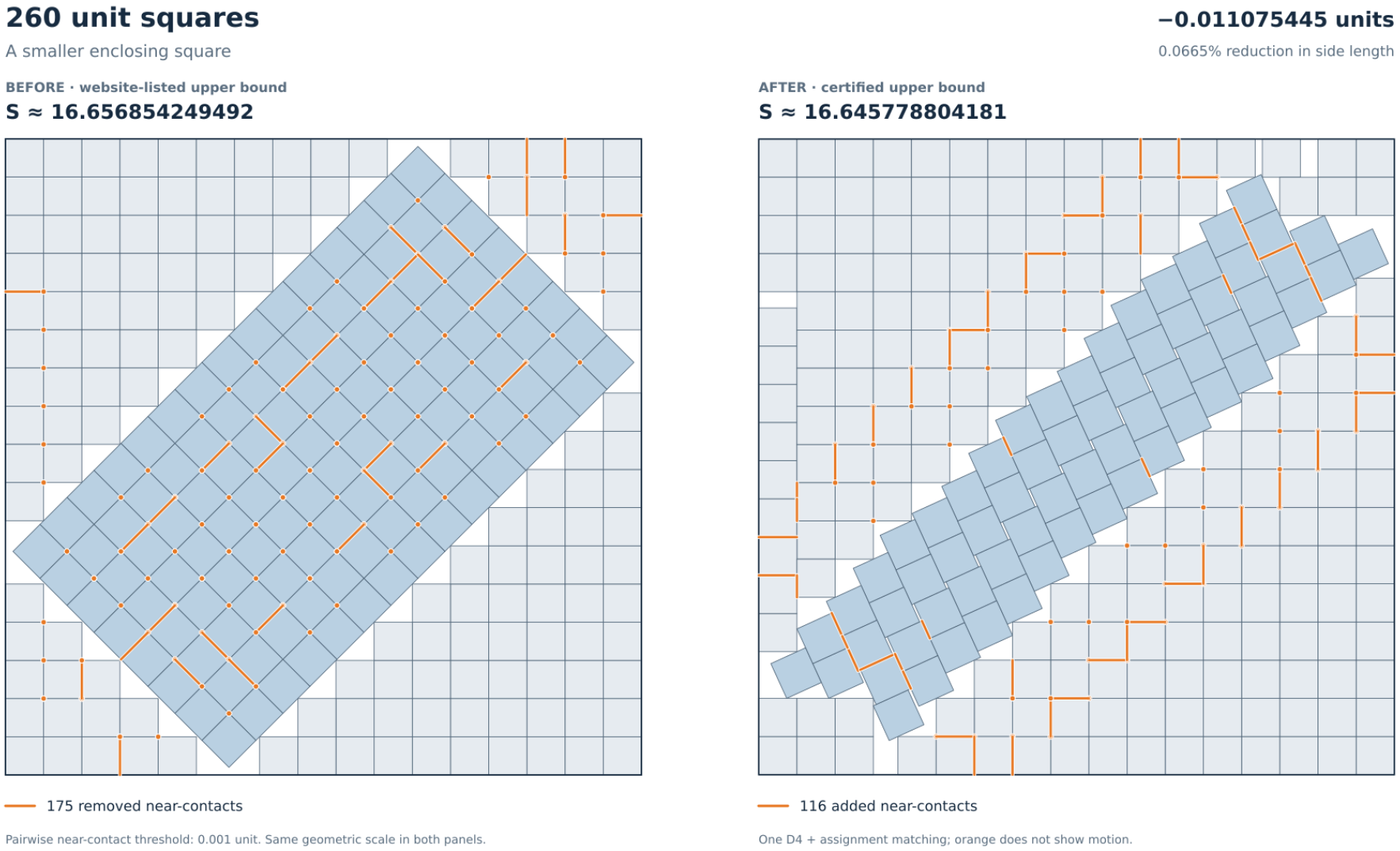
Website 15.656854249492 -> Certified upper bound 15.650174756949



Orange identifies changed near-contact adjacencies after the recorded square-symmetry and tile matching. The accompanying topology JSON states the tolerance and correspondence; the exact certificate, not the figure, establishes feasibility.

260 unit squares: before and after

Website 16.656854249492 -> Certified upper bound 16.645778804182



Orange identifies changed near-contact adjacencies after the recorded square-symmetry and tile matching. The accompanying topology JSON states the tolerance and correspondence; the exact certificate, not the figure, establishes feasibility.